



Hands on Homework #1 (Non-parallel Axis)

EGN 3310 - 0M01 Engineering Analysis: STATICS

MWF, 11:30AM-12:20AM

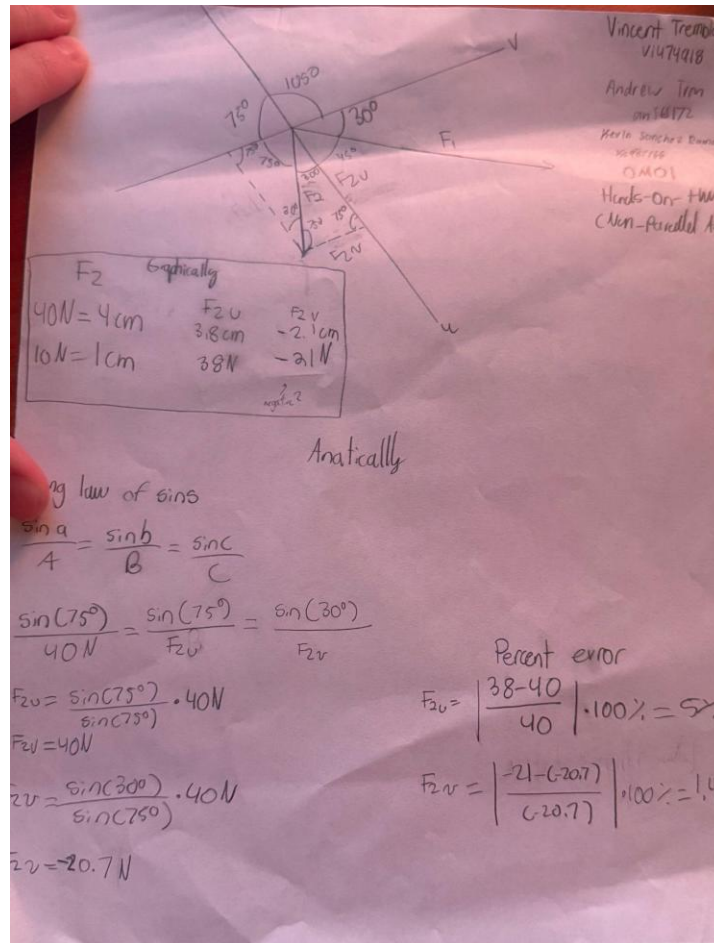
Submitted by: Group 38

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Graphical and Analytical Worked Out on Paper



Hands-On Method (Physical Model)

In this first hands on homework, the physical model was done by graphing out a problem similar to the one that was given.

To do this, accurate measurements were required, so the use of a ruler and protractor was necessary.

After drawing out the lines with their according angles, we were able to use the fact that the axes were non-perpendicular to draw parallelograms. This made it possible to find all the remaining angles we were missing to solve the problem.

Given the fact that F2 was 40N, we made a scale of 10N=1cm so that we were able to match force with the physical length of the vector on paper. Following that scale, we split F2 into F2u and F2v and measured their lengths as well, which we calculated as:

F2u	3.8 cm	38N
F2v	-2.1cm	-21N

This was the final measurements we needed to complete what was asked of us, so that concludes the Hands-On method.

Hand Calculation (Theoretical Method)

For the calculations we utilized the graph we used for the hands on method, as it contained the angles needed to complete this.

As I previously mentioned, the axes are not perpendicular. This means that using a right triangle and using sin and cos is not a viable option. This meant we had to use the law of sines in order to figure out the F2 into **u** and **v**.

The law of sines state that: $\sin(a)/A = \sin(b)/B = \sin(c)/C$

Using that and the fact that we had every angle as well as A, which we set as 40N as it is the length of F2, we were able to solve for B and C.

The angle opposite to F2 is 75°. That means we have all the components to solve for B as we have the other angles.

Here is the formula to solve for B: $\sin(75^\circ)/40N = \sin(75^\circ)/F2u$

$$F2u = 40N * \sin(75^\circ)/\sin(75^\circ)$$

Solving that we get that F2u= 40N.

This means that both F2 and F2u are 40N, which makes sense as we are working with an isosceles triangle.

Now we can solve for F2v using the exact same method

Here is the formula to solve for F_{2v} : $\sin(75^\circ)/40N = \sin(30^\circ)/F_{2v}$
 $F_{2v} = 40N * \sin(30^\circ)/\sin(75^\circ)$

Solving for that we get $F_{2v} = -20.7N$

That concludes the hand calculation method, as we now have both the measurements of the **u** and **v** components of F_2 .

Conclusions and discussions

Vincent Tremblay (vi474918)

Overall, the homework went really smoothly. We were able to properly draw out the graph to have an semi- accurate measurement of each line and apply concepts for the hand calculation portion as well.

Looking at the data itself, we can see that previous mathematical concepts show that our results are correct.

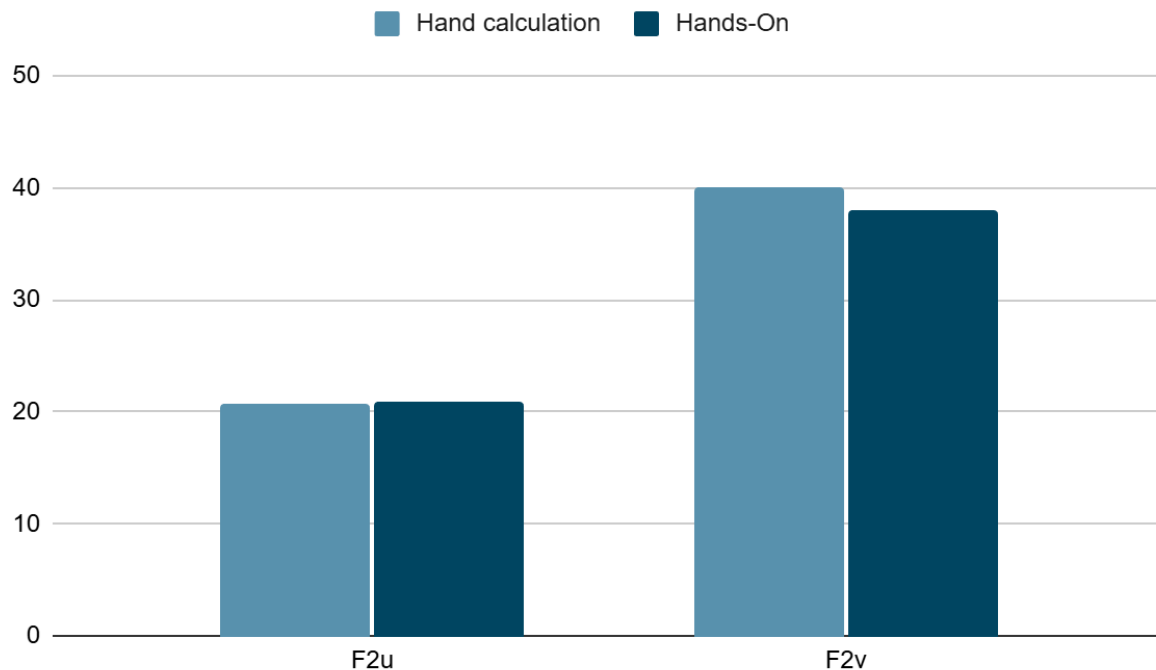
For example the fact that we have an isosceles triangle means that two sides should be the same, and indeed they were.

Probably the hardest part of the lab was drawing the graph as accurately as possible since using the conversion $1cm=10N$ meant that we needed as accurate measurements as possible for the hands on homework.

Comparing the data for both, here is what was obtained

	Hands-On	Hand calculation
F_{2u}	38N	40N
F_{2v}	-21N	-20.7N

Hands-On VS Hand-calculated F2u and F2v (Newtons)



The graph above showcases F2 in its components in Newtons. For the hand's on (dark blue line), 1cm = 10N

From this comparison, it can be seen that the Hands-On method is not ideal, and contains some reasons (although it is quite close).

It is great to have hands-on data however as it is a good reference point to see if your hand calculations are in the right area.

In terms of % error, here are the formulas with the data we obtained. The hand calculation data is used as the actual value as it is the most accurate.

$$F2u = (|38N - 40N| / |40N|) * 100\% = 5\%$$

$$F2v = (|(-21N) - (-20.7)| / |(-20.7)|) * 100\% = 1.45\%$$

Looking at this percent error, our hands-on calculations were not too far off, which is great. However this only reinforces the fact that there will always be some error with hands on calculations, as there will always be a slight degree of inaccuracy, which is expected.

Andrew Tran (an365172)

After obtaining the results of F2u and F2v using both the analytical and graphical approach, we saw that they were similar to each other. This is evidence of the Law of Sines, as shown when applying the concept to find F2u and F2v analytically. Finding F2u also showed evidence of the Isosceles Triangle; if 2 angles in a triangle are the same, so are the lengths of the sides across each angle.

	Graphical	Analytical
F2u	38N	40N
F2v	-21N	-20.7N

The graphical approach had slightly different answers from the analytical approach because we were tasked to copy 2 nonperpendicular axes as well as the magnitude of force F2. We had to draw F2 to the ratio of 10 centimeters per newton (10 cm/N). In this case, we drew F2 to 4 centimeters since F2 was 40 Newtons. This produced expected inaccuracies and thus errors in our result from the graphical approach vs analytical.

By using the formula to calculate percent margin of error:

$$\% \text{ percent error} = \left| \frac{\text{result} - \text{expected} - \text{result}}{\text{expected result}} \right| * 100$$

This was was our percent error for each force component:

$$F2u = \left| \frac{38 - 40}{40} \right| * 100 = 5\%$$

$$F2v = \left| \frac{-21 - (-20.7)}{-20.7} \right| * 100 = 1.45\%$$

Considering that the percent error was only 5% for F2u and 1.45% F2v, I believe that we did a good job producing data close to the actual results (which was the analytical approach). Regardless, there will always be some error in experiments where humans (us) have to replicate a model by hand.

Kevin Sanchez Ramos (ke987766)

Using graphical and analytical approaches to measure the forces in a system helped demonstrate how accurately real-world situations can be modeled. In our Hands-On-HW 1 assignment, by drawing the vectors to scale and solving for the unknown forces, we found that both results ended up being somewhat similar.

For instance, using a scale conversion of 1 cm = 10 N allowed us to reduce the margin of error and calculate the forces analytically with ease, as we used the law of sines to solve for all the unknown variables. However, despite our attempt to draw a perfect measurement graphically, we would end up making a margin of error, as shown in our final result.

Given:

Calculations results:	Hands-On	Hand calculation
F2u	38N	40N
F2v	-21N	-20.7N

Ultimately, calculating for our percent error:

$$F2u = \frac{|38N - 40N|}{|40N|} \times 100\% = 5\%$$

$$F2v = \frac{|(-21N) - (-20.7N)|}{|(-20.7N)|} \times 100\% = 1.45\%$$

All things considered, compared to our final numbers and seeing our percent error, we came very close to the exact number despite human error. In addition, the fact that both numbers came out very close shows that we were also heading in the right direction in terms of our calculations. However, because of human error, we are shown that no matter how accurate we believe something to be, there will always be a tiny margin of error in a design or calculation, despite our best attempts.